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CONSERVING WATER WITH A PLC BASED IRRIGATION SYSTEM

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ABSTRACT

Farming accounts for about 70% of the freshwater consumption. The existing timer-based approaches utilized in farming results in waste of significant amount of fresh water. This paper details the development of a PLC based automated irrigation system which can adapt the daily irrigation amount according to the varying climate and type of crops. The system uses simple sensors to adapt daily irrigation amounts to plant needs. The water levels can be directly sensed by PLCs which then adapt the irrigation schedule to the observed conditions, saving great amounts of water.

Keywords: Automation, Instruction List, Ladder Diagram, PLC, Water conservation.

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1. INTRODUCTION

Water is a precious natural resource and is critical for agriculture, industries, and domestic use. Farming accounts for 70 percent of the water consumed in most regions of the world [1]. The use of simple controllers that use fixed schedule and fixed run times to operate irrigation systems, regardless of weather conditions leads to wasting of water. In addition to flooding of farms, the less known adversities are water pollution from nutrients, pesticides and other contaminants which can lead to economic and environmental costs [2].

The recent technology in irrigation is prone to certain flaws resulting in over irrigation, wastage of water, and power [3]. Hence, there is a need for more efficient controllers. The purpose of this work is to implement an automated irrigation system by sensing the water level in the fields depending on the type of crops. The substantial development of Programmable Logic Controllers (PLC) and their affordable price makes them suitable for use in irrigation systems. PLC's are presently being used extensively in manufacturing processes. They have a processor which inherits the capacity to command a number of electric devices through relays. PLCs are immune to electrical noise, resistant to vibration, and are designed to operate under extended temperature ranges.

The aim of this work is to develop an economical PLC based system that automatically adapts to actual weather conditions, using simple criteria, and then executes the irrigation accordingly. This system can be mass produced and adopted by farmers, municipalities, and companies in any country where irrigation is needed. In this work, an automated irrigation system is implemented using Programmable Logic Controller with the aid of sensors and solenoid valves to conserve water and power. The resulting system would be highly efficient, reliable, cost effective, easy to control and sustainable to rigorous weather change.

2. SYSTEM BLOCK DIAGRAM

The block diagram of the automated irrigation system is shown in Fig 1. PLC is the central part of this system. The setup utilizes water level sensors (S1-S4) in the fields and the signals are fed back to the PLC which play a key role in the automated irrigation system. Sensors can also be placed in the water sources for the protection of pumps from cavitation. The water level sensors can be resistive, capacitive, admittance-type or magneto resistive.

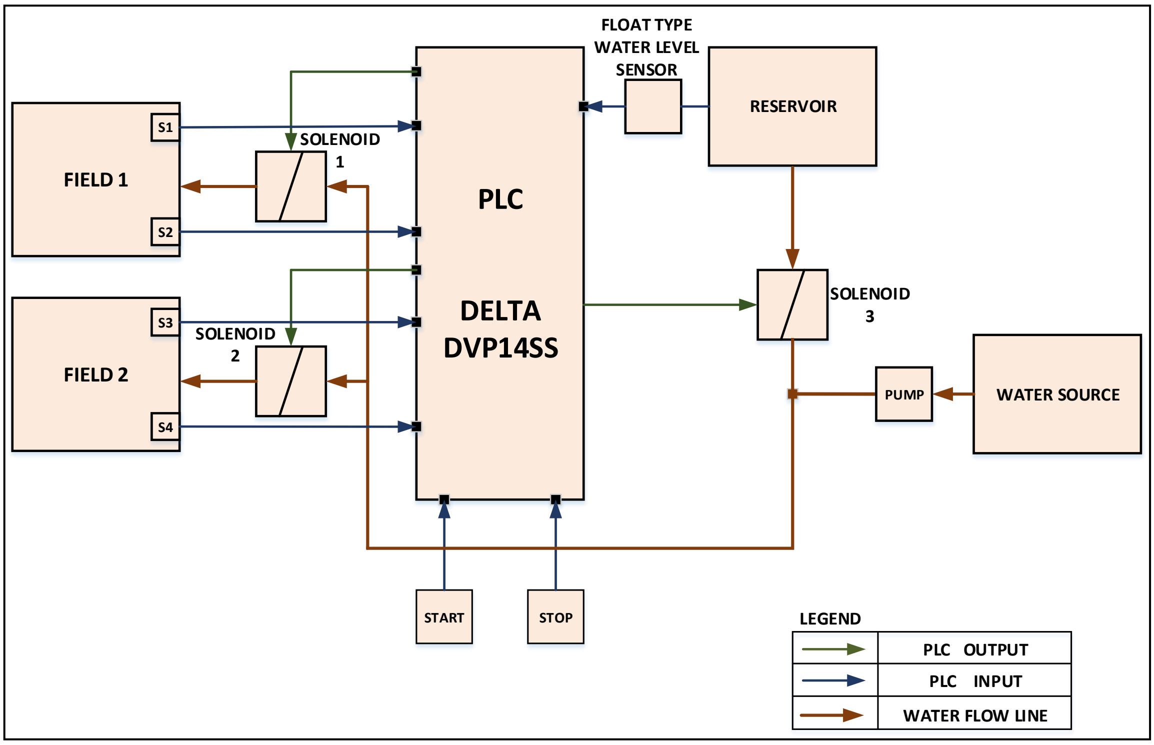


Figure 1: System block diagram

Pumps and solenoid valves are used to automate the flow of water, remotely by the signal from the PLC. This setup provides provision for the usage of alternative water sources to demonstrate the flexibility of PLC based system.

3. THEORY OF OPERATION

The START button enables the irrigation system and the STOP button disables the irrigation system for maintenance or during non-operation, saving power. When the sensors in the field (S1 – S4) indicate low water level/low moisture content, PLC actuates flow of water from water sources depending on their levels. Under normal condition, the PLC drives the water from overhead reservoir (Water source 1) to the fields and enables solenoids 1-3. However, if the level of water in the reservoir goes below the predetermined level, PLC disables solenoid 3. Thus, PLC switches its control to the water source 2 by turning ON the pump.

If the level of water in the water source 2 goes below the predetermined level, the PLC turns OFF the pump to prevent cavitation. Multiple sensors can be used in the field to reduce errors due to unevenness of the land. The logic can be implemented to begin irrigation by ANDing or ORing all the field sensor feedback. When the water level equals or rises above the set value, the sensor sends signal to PLC to turn OFF or close the valve. Thus, by the application of PLC, both automation of irrigation and prevention of under & over irrigation can be achieved. Unlike timer-based controllers, PLC continuously monitors level sensors and irrigates the land when needed.

4. PLC PROGRAMMING

PLCs are intended to be used with minimal programming effort. Several PLC programming languages are used which can then be converted into machine code by some software for use by the PLCs. To standardize PLC programming languages among PLC manufacturers & users, the IEC 1131-3 was published, and following languages were specified: Ladder Diagrams, Instruction List, Sequential Function Charts, Structured Text, and Function Block Diagrams. [6]

4.1. PLC Ladder Programming (LAD)

Ladder diagrams are very commonly used method of programming PLCs. In ladder diagram programming, the two vertical lines represent the power rails. Circuits are connected on the horizontal lines, i.e. the rungs of the ladder. One operation is defined per rung of the ladder in the control process. The ladder diagram is read from top to bottom and from left to right. When the PLC is in run mode, it goes through the entire ladder program to the end, and then resumes at the start as shown in Fig 2 [6].

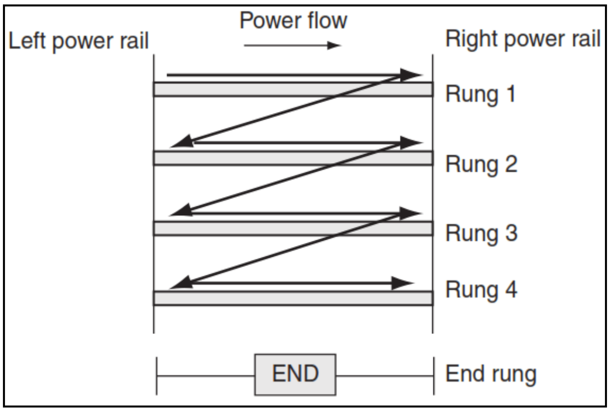
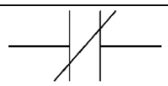
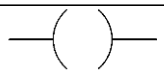


Figure 2: scanning motion employed by the PLC

Each rung starts with at least one input and must end with at least one output. The electrical devices are represented in their normal condition. A switch that is normally closed is shown closed and a normally open switch is shown as open as shown in Table 1 [6]. The devices can appear in more than one rung of a ladder. The notation for the devices is specific to the PLC manufacturer. The inputs and outputs are identified by their addresses and these addresses are to the memory locations of the PLC. The ladder diagram code for the automated irrigation algorithm is shown in Fig. 3.

Table 1: Ladder diagram basic symbols

Symbol	Description
	Normally Open (NO) contacts (Switch, relay or ON/OFF devices)
	Normally Closed (NC) contact (Switch, relay etc.)
	Coil (output loads: Solenoid, motor/pump)

4.2. Instruction List (IL)

Instruction List programming methodology can be considered to be entering of a ladder diagram using text. Instruction list is a series of instructions, each instruction being on a new line and is similar to the assembly language. An instruction has an operator (command) followed by at least one operand (variable, constant or instance name) and one optional modifier, respectively. Instruction list is a fundamental programming language and other PLC programs can be converted to IL. The basic operators, its operation and its ladder diagram interpretation is shown in Table 2 [6]. The Instruction List code for the automated irrigation algorithm is provided in section 5.2.

Table 2: Instruction List - Operator & Operation Table

IEC 1131-3 Operators	Operation	Ladder diagram
LD	Load operand into result register	Start a rung with open contacts
LDI	Load negative operand into result register	Start a rung with closed contacts
AND	Boolean AND	A series element with open contacts
ANI	Boolean AND with negative operand	A series element with closed contacts
OR	Boolean OR	A parallel element with open contacts
ORI	Boolean OR operand with negative operand	A parallel element with closed contacts
OUT	Store result register into operand	An output

5. ALGORITHM

In this work the algorithm has been implemented in Ladder diagram (LAD) and Instruction List (IL) and are described below.

5.1. Ladder diagram implementation:

The Ladder Diagram (LAD) implementation of the automated irrigation system algorithm is shown in Fig. 3. The field sensors in this implementation send a HIGH signal when water is below the sensor and LOW signal when water is above the sensor. In the first rung, sensors of both fields are given as input (I2 – I5) through Normally closed (NC) switches to the PLC in 'OR' Logic.

When the START switch is ON and if any one sensor is LOW, the latch relay Q1 gets latched. Once the Q1 is latched, the next step is to determine the water source by activating/deactivating respective solenoid valves and pumps.

In the second rung the overhead Reservoir sensor (I5) is used as input which controls the TIMER (T0). Normally closed switch is used for this purpose since the sensor is placed at the bottom of the reservoir, which sends HIGH signal when water goes below the sensor.

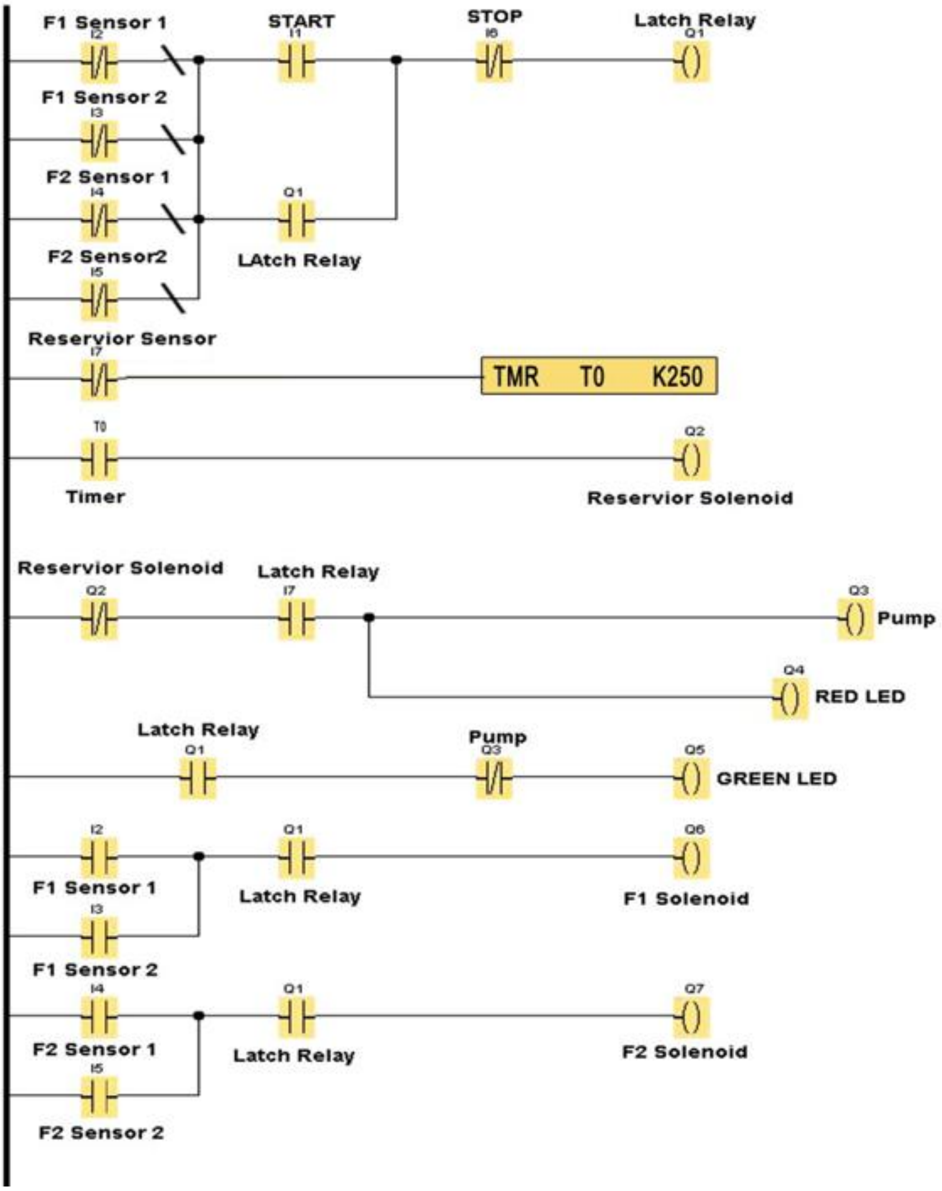


Figure 3: Ladder diagram (LAD) implementation of Algorithm

Reservoir solenoid (Q2) in the third rung turns ON after a user specified delay of K units depending on the time base of the PLC. Here the Constant K = 250 with a time base of scan time. And when level of reservoir is low, pump (Q3) is turned ON and control transfers to water source 2 as shown in fourth rung. The Timer unit is used to prevent toggling of control from water source 2 to reservoir and vice versa when level changes in the reservoir. The fifth rung displays the pump status by green LED (Q5).

In the sixth rung the field solenoids are controlled by the field sensors and latch coils. Each field has two sensors in OR logic with Normaly Open switches. Any number of sensors can be added to improve the accuracy. When there is water demand the switches gets closed, thus sending HIGH signal to PLC to activate Solenoid valves of the respective fields. When all the sensors are ON, both the solenoids will be ON. When there is no demand for water the system is OFF thus saving power. To stop the system, a STOP switch (I5) in normally closed position is used in AND logic with START in the first rung.

5.2. Instruction list implementation

000000	LDI	S2	000011	LD	S1	000025	OUT	Y3
000001	INV		000012	OR	Y0	000026	LD	S2
000002	LDI	S3	000013	ANB		000027	OR	S3
000003	INV		000014	ANI	S6	000028	AND	Y0
000004	ORB		000015	OUT	Y0	000029	OUT	Y4
000005	LDI	S4	000016	LD	S7	000030	LD	S4
000006	INV		000017	TMR	T0 K250	000031	OR	S5
000007	ORB		000021	LD	T0	000032	AND	Y0
000008	LDI	S5	000022	OUT	Y1	000033	OUT	Y5
000009	INV		000023	LDI	S7	000034	END	
000010	ORB		000024	AND	Y0			

SIMULATION

To validate the algorithm, a PLC simulation tool, Delta WPL Soft was used. The simulation tool saves time and also increase the level of safety by testing the effectiveness of algorithm before the system is activated. DELTA WPLSoft is a program editor for Delta DVP series PLCs for WINDOWS computers. It supports Ladder diagrams, Sequential Function Chart, Instruction List and also enables communication between PLC and PC [8]. ON state of each component is highlighted by green color. Several test cases were simulated, and the algorithm was verified. An example of one of the simulation test cases is shown in Fig 4 - 10.

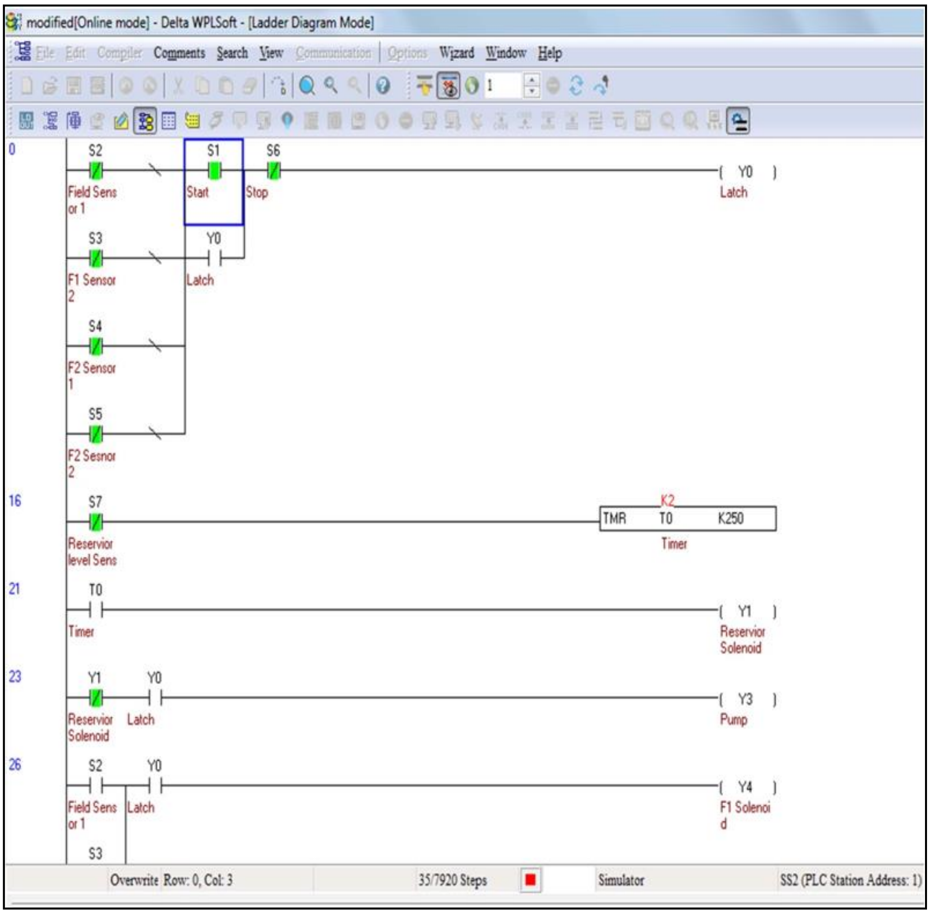


Figure 4: Simulation: Y0 latch is not enabled when sensors indicate no water demand

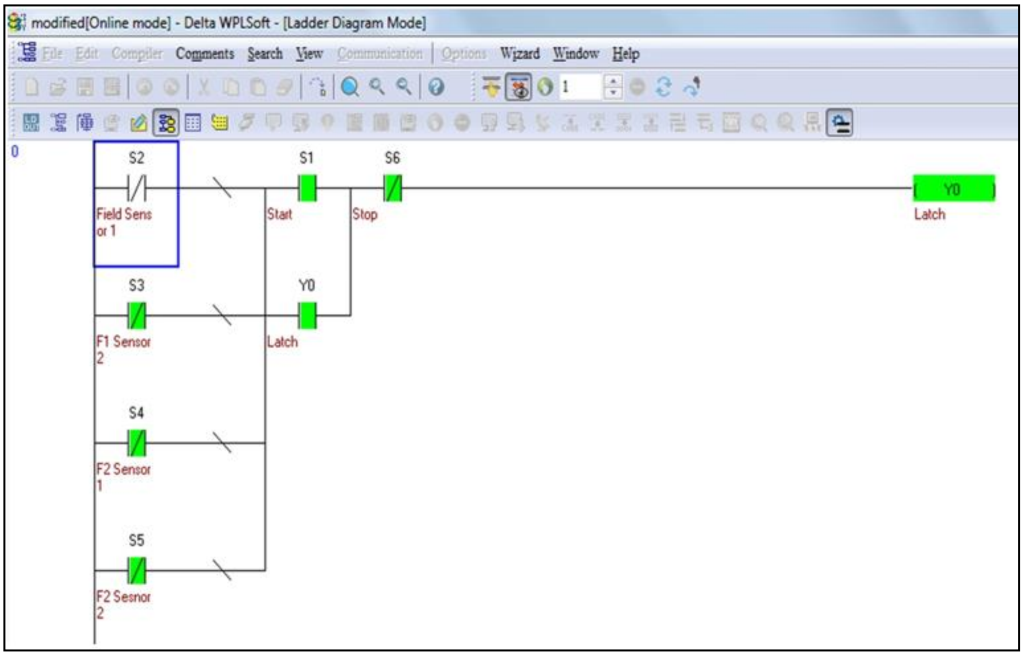


Figure 5: Y0 latched with one of the field sensors (S2) indicating water demand

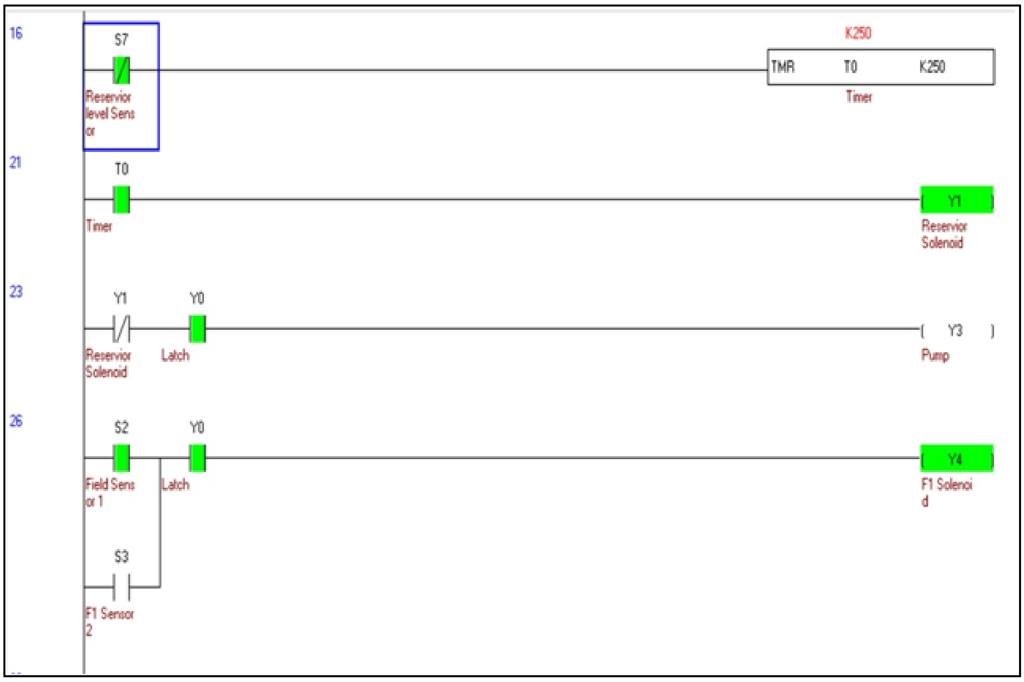


Figure 6: Field-1 Solenoid-1 is enabled and water flows from Reservoir

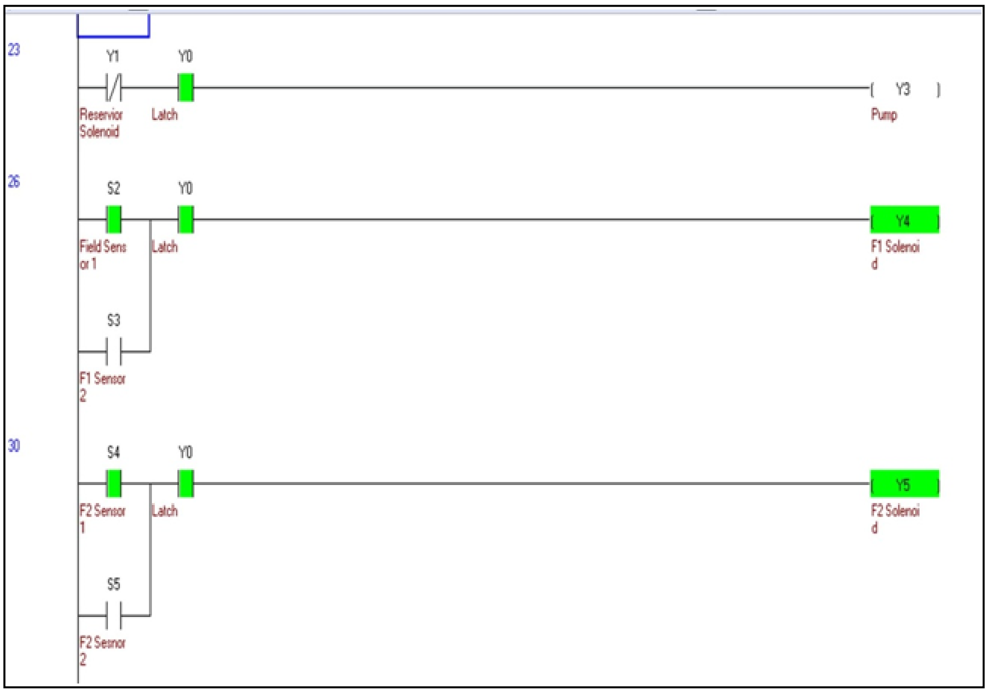


Figure 7: When Field 2 sensor also indicates water demand, Solenoid 2 is enabled

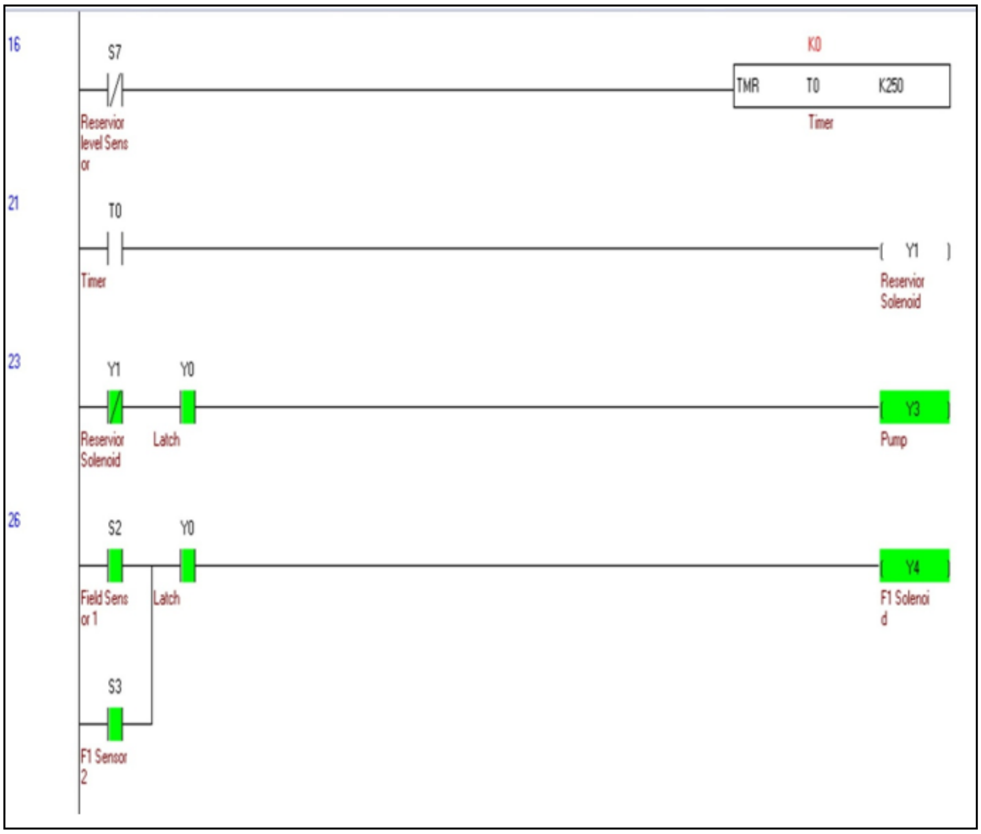


Figure 8: Reservoir level is low, Solenoid 3 is disabled, and water source 2 pump is enabled

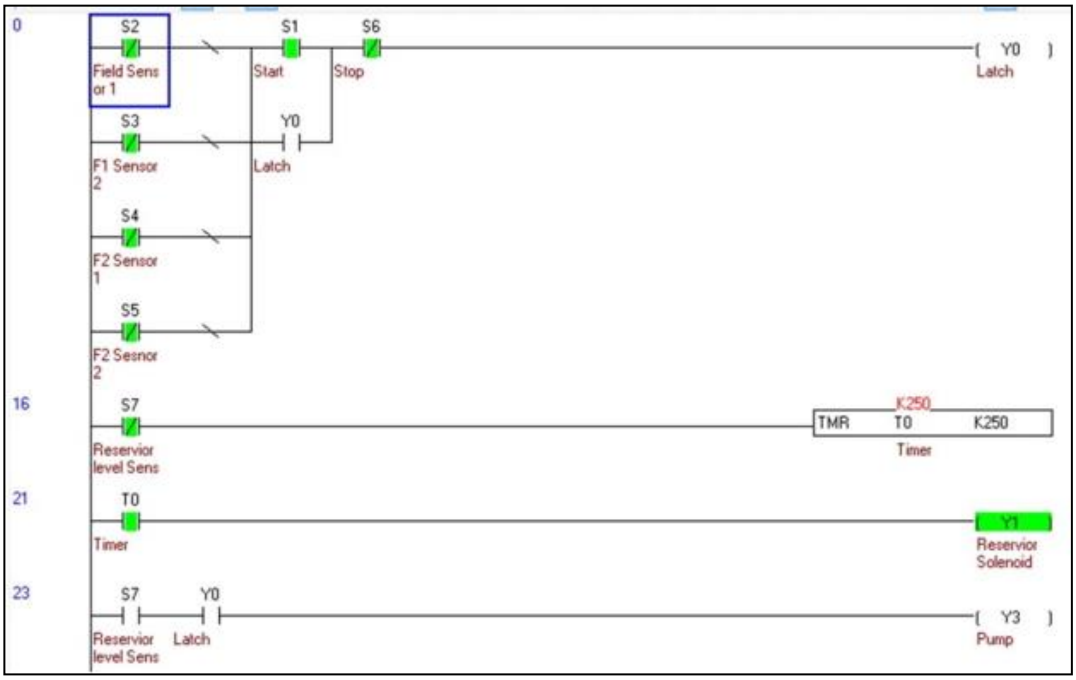


Figure 9: All field Sensors indicating no water demand, Y0 latch is disengaged

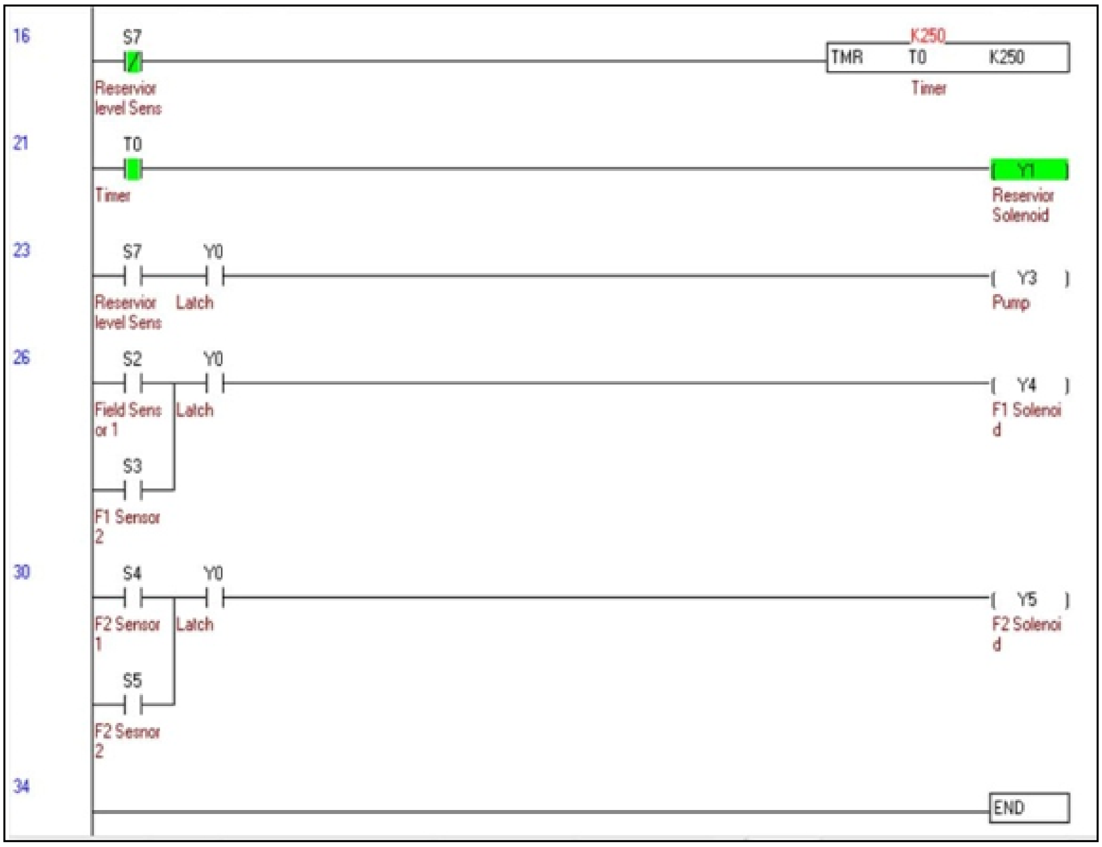


Figure 10: All solenoids are disabled, and irrigation has stopped

6. PROTOTYPE IMPLEMENTATION

In this setup, Delta DVP-14SS2 series of PLC is used. It has 8 digital inputs and 6 digital output ports with maximum input/output ports extension up to 128 ports. The PLC has 8000 steps of programming capacity and has enhanced real-time monitoring capability. The compact design lets the DVP14SS2 controller thrive in limited panel space. It makes use of WPLSoft Software for coding ladder diagram and has a built-in RS-232/RS-485 ports for communication with PC. The software converts the Ladder diagram to Instruction List and Functional Block Diagram.

Figure 11: Delta DVP-14SS2 PLC

A submersible pump of 16W has been used in the prototype setup. A normally closed Solenoid valve rated at 24V, 500mA, 2-way direct acting valve was used. The solenoid operates well at 11V, 450mA and the power supply has been implemented appropriately. The Solenoid valve coil which is normally in de-energized state keeps the plunger closed thereby preventing the flow of water through it. The coil is energized by PLC on monitoring sensors and energizing solenoid to enable irrigation.

Figure 12: Solenoid Valves

Water level sensor has been implemented using the partial conductivity property of water, instead of readily available off the shelf sensors. The construction of the sensor is shown in Fig. 13. A pair of electrode probe with 18V is placed at the bottom of the tank. When the water level is above the level of the probe, it forms a closed path resulting in near zero voltage (LOW) read by the PLC. Once the water drops below these probes, the voltage read by the PLC will be 18V (HIGH). The PLC detects >16V as HIGH and <14V as LOW. 230Vac is stepped down with a step-down transformer, followed by a diode bridge rectifier, capacitor filter and a linear voltage regulator IC 7818 to produce 18VDC.

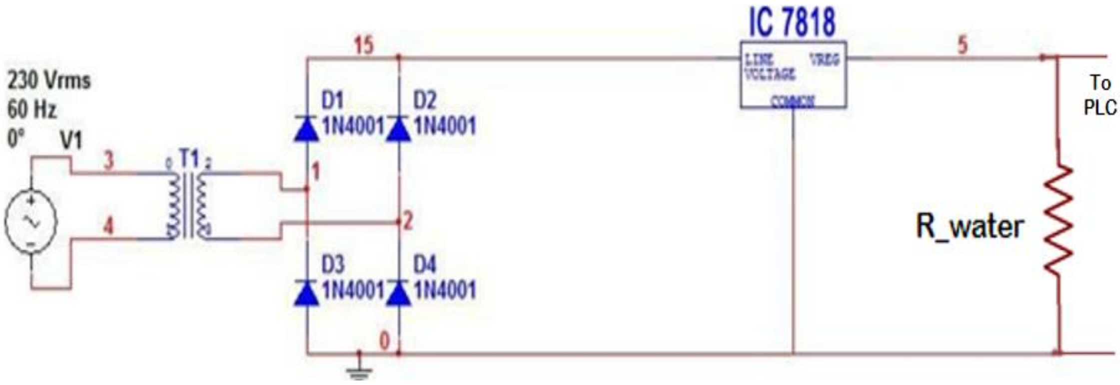


Figure 13: Water level sensor

The 230Vac is stepped down using a transformer and rectified with a diode bridge rectifier. From the rectified output several DC bus voltages have been generated as shown in Fig. 14: 24V to PLC, 12V to Solenoid valve, 18V for sensors and 12V for the relay.

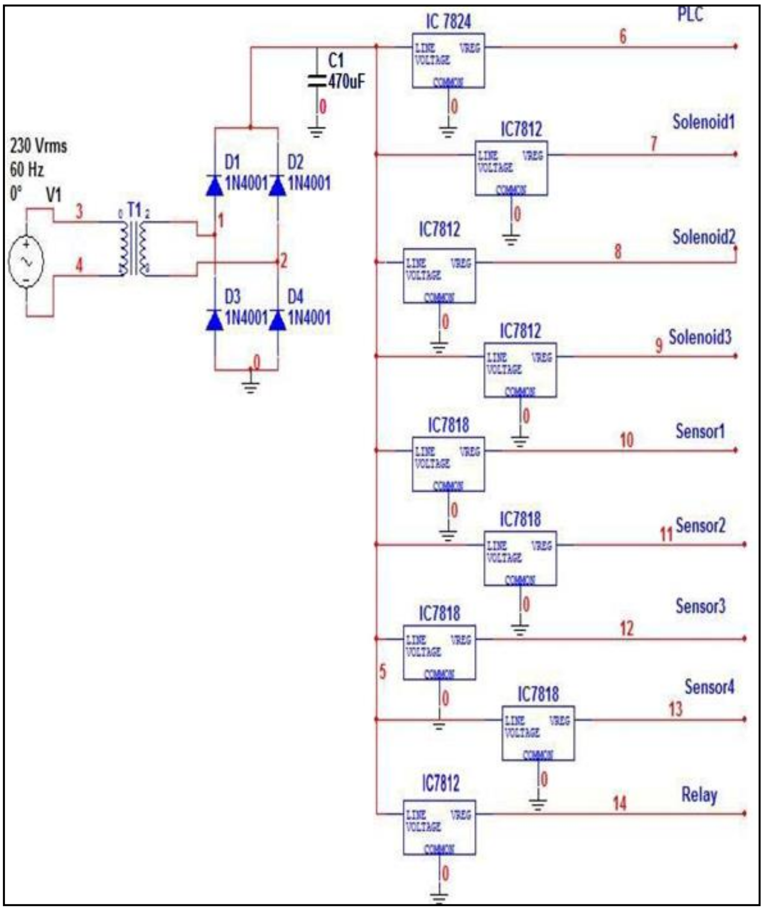


Figure 14: Power supply circuit

The final setup of the PLC based automated irrigation system is shown in Fig. 15.

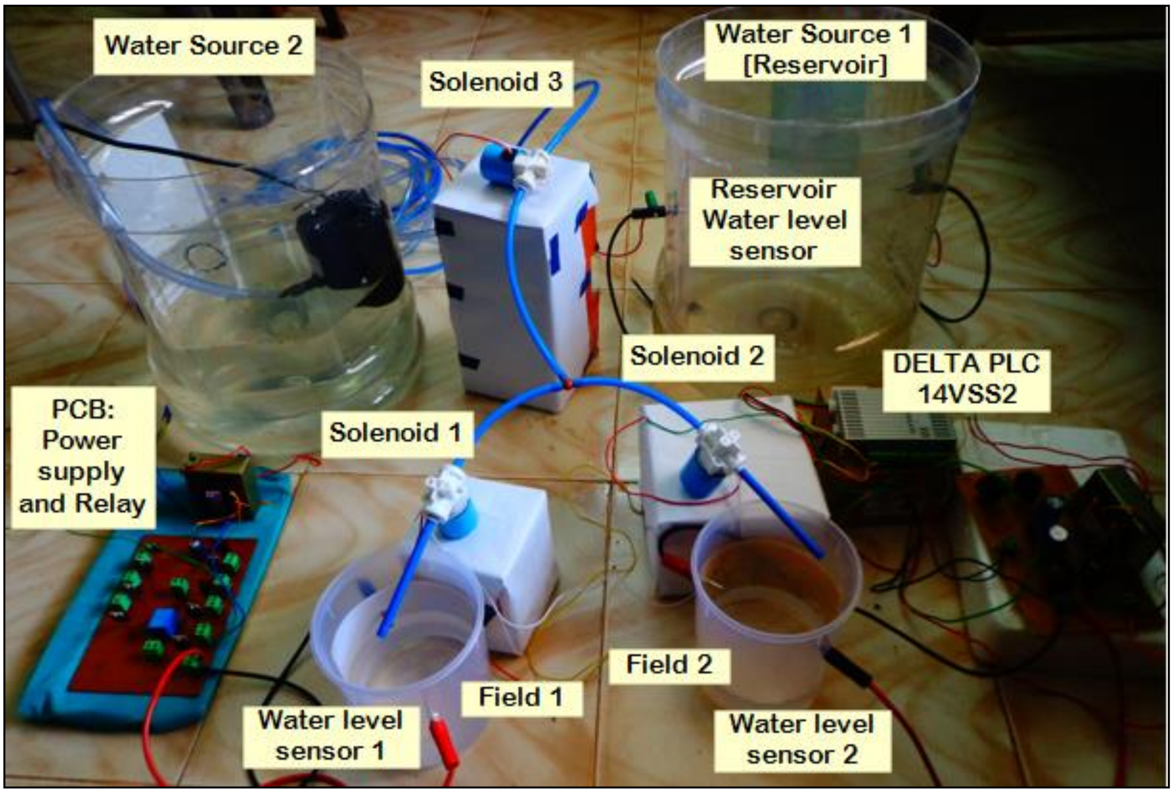


Figure 15: Final setup of PLC based Automated Irrigation System

7. OBSERVATION & RAMIFICATIONS

The algorithm has been verified through simulation and prototype testing. The prototype as shown in Fig. 15 operates per design and matches with the simulation. This proof of concept can be scaled to implement the irrigation system on an actual field with off the shelf components for higher reliability with better packaging and accuracy. The linear regulator power supply section can be replaced with switching converters to improve efficiency.

Several sensors like temperature, timer, water/moisture sensors, GPS and weather forecast can be integrated to the system to form a more intelligent algorithm to further optimize the system. A graphical user interface (GUI) can be implemented to modify certain thresholds in the algorithm to customize per use case, thus making it a universal product. The working principle can be extended to applications like sprinkler systems in domestic and municipality use.

8. CONCLUSIONS

The PLC based automated irrigation system has been verified through simulation and prototype testing and has demonstrated to be a potential replacement for timer-based irrigation systems. This system through a proof of concept has demonstrated potential to be a solution for conserving water and power. The working principle has scope for various other applications leading to water and power conservation with minimum investment and maintenance.

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